

Implant Surface Innovations: What Really Improves Integration?

Group 6

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Behavioral Objectives

1. Describe the biological cascade of osseointegration, including protein adsorption, cell adhesion, osteoblast/osteoclast activity, and biofilm interactions at the implant-tissue interface.
2. Explain how surface energy, topography, roughness, and wettability influence osseointegration, peri-implant tissue health, and biofilm adhesion.
3. Differentiate the surface modifications of titanium implants (machined, sandblasted, acid-etched, anodized, hydroxyapatite-coated, nanostructured, bioactive coatings, and others) by explaining their influence on bone healing, cell attachment, and microbial colonization.
4. Compare and contrast the clinical outcomes and limitations of implant surface innovations based on scientific evidence, focusing on osseointegration rates, peri-implant stability, and risk of peri-implantitis.
5. Apply a biologically informed decision tree to select implant surfaces and adjunctive treatments tailored to patient-specific factors such as bone quality, systemic health, microbial risk, and esthetic demands.

Behavioral Objective #1

Describe the biological cascade of osseointegration, including protein adsorption, cell adhesion, osteoblast/osteoclast activity, and biofilm interactions at the implant–tissue interface.

Katlyn Williams and Nathalie Paladines

Search Strategy



- P:** Patient's receiving dental implants
- I:** Placement of titanium or surface-modified dental implants
- C:** Early vs. late stages of osseointegration
- O:** Successful Osseointegration

Databases Used

PubMed

Keywords

Dental implants, osseointegration, implant-tissue surface, biological cascade,

PubMed search string

(dental implants) AND (osseointegration) OR (implant-tissue surface) OR (biological cascade)

Search results

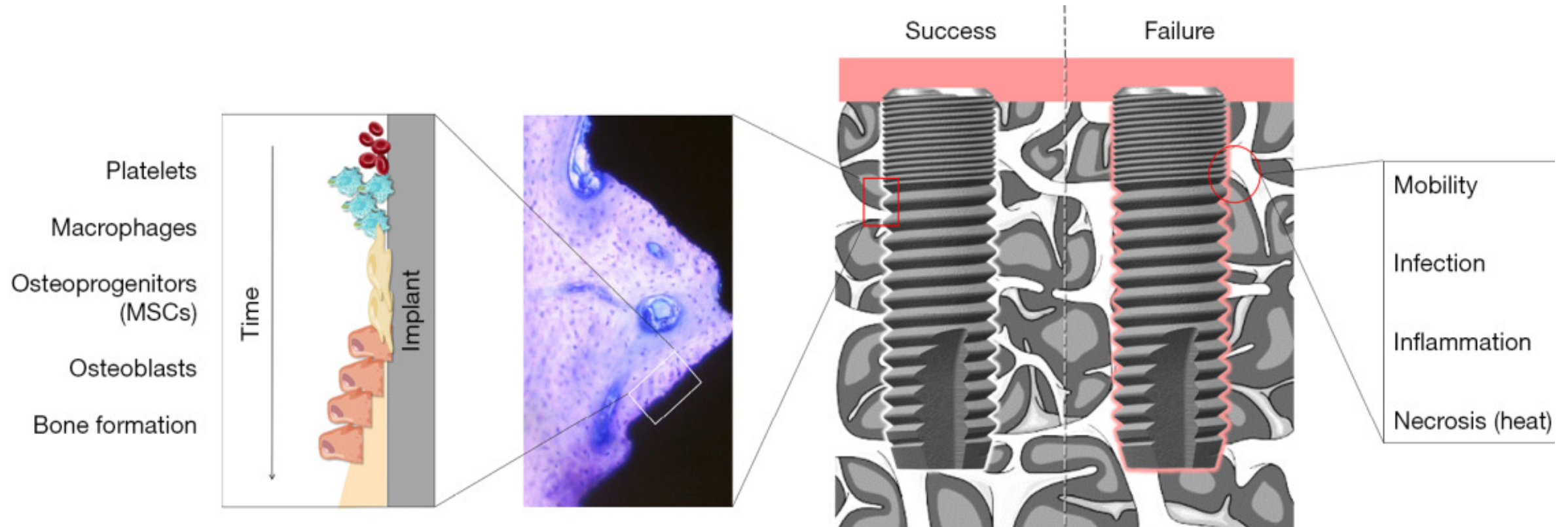
5231 results, 3 used

Highest level of evidence

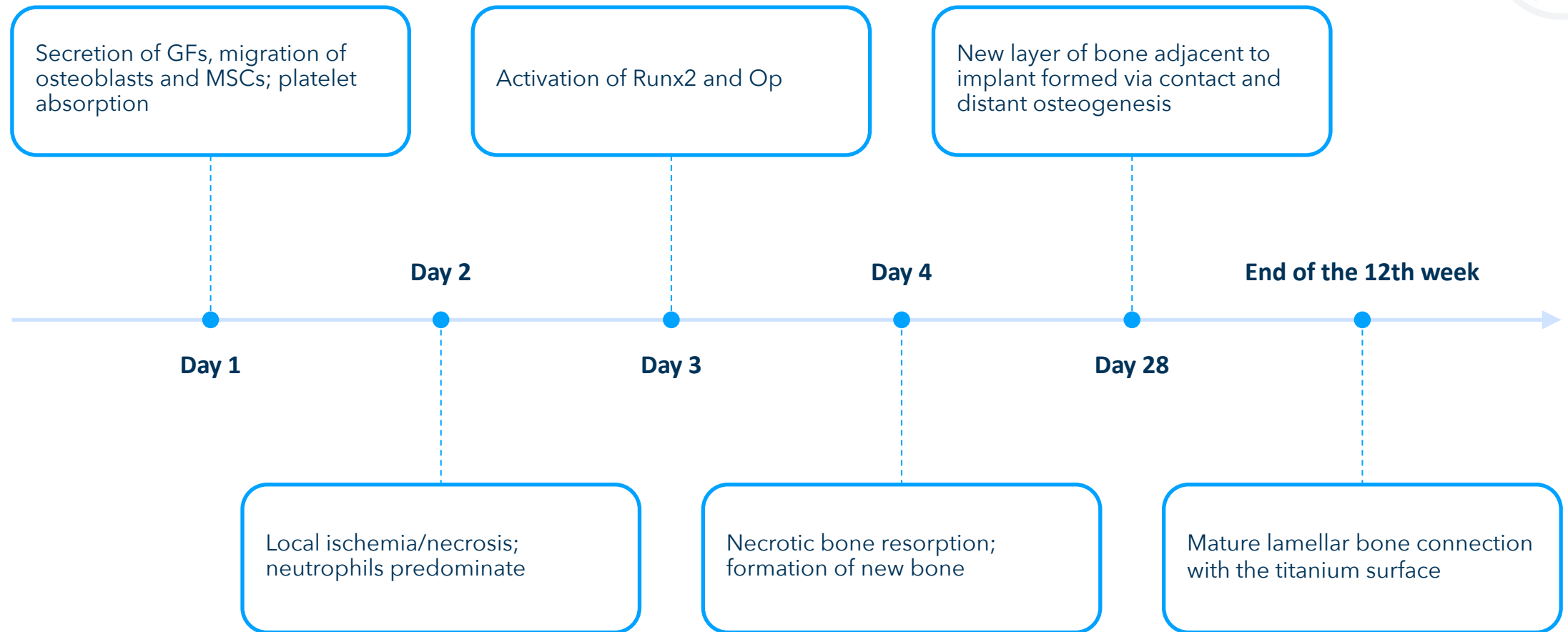
Systematic Review and Meta Analysis



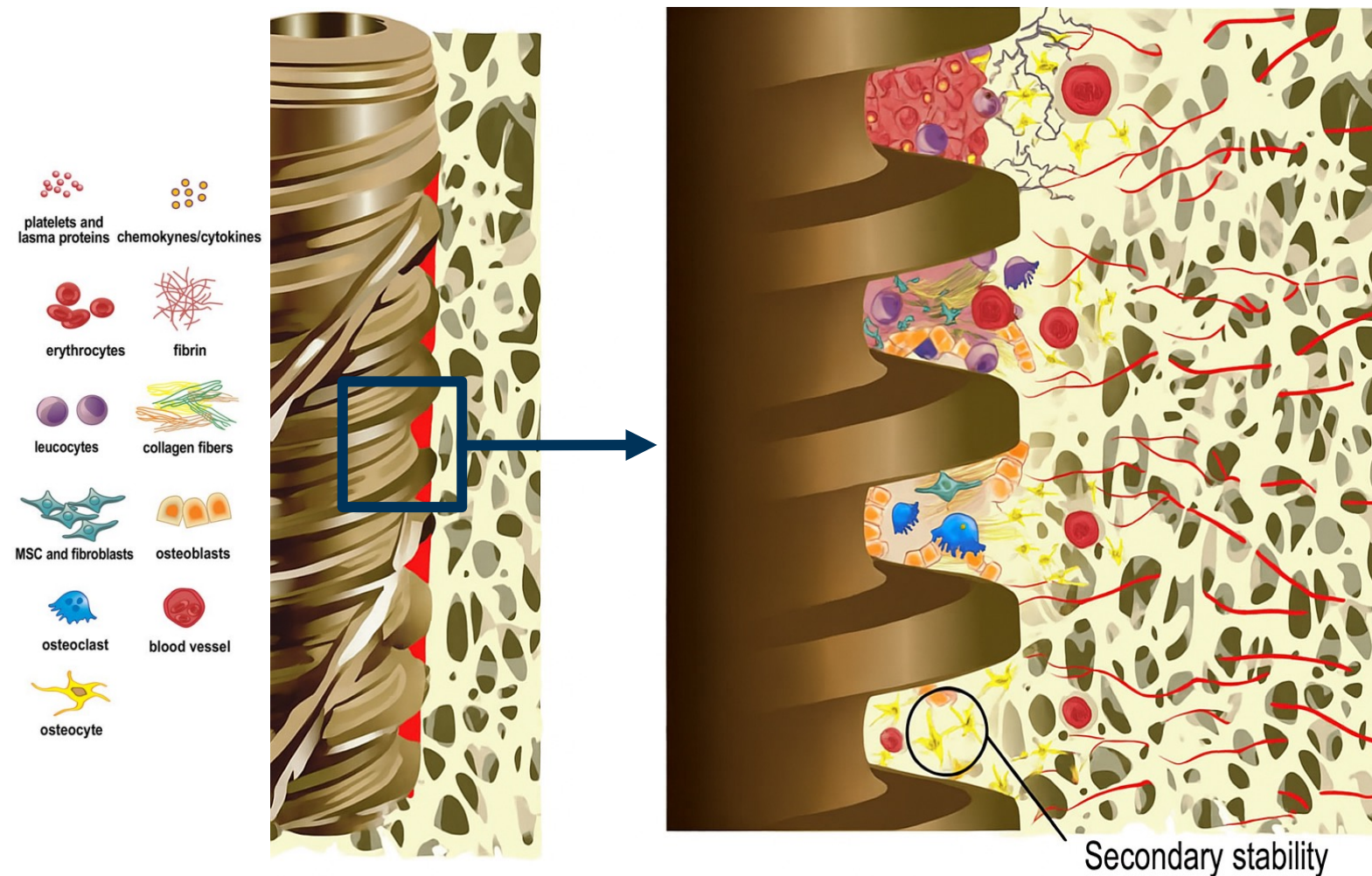
The Biological Cascade of Osseointegration



Cellular and Molecular Interactions



GFs: Growth Factors; MSCs: Mesenchymal Stem Cells



Day 1

- GF secretion and protein adsorption
- Cell adhesion, migration, and differentiation
- Inflammatory cell adhesion

Day 7

- Activation of Runx2 and Op TFs
- Angiogenesis
- Recognizable bone matrix cohesion on surface of implant

Day 16

- Implant surface entirely coated with mineralized tissues, osteoid, and dense matrix

Day 28

- Complete binding of bone along implant surface
- Tissue layer of collagen and osteoblasts
- Bone regeneration

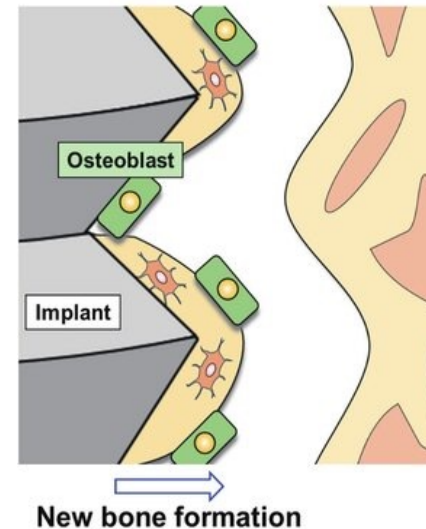


The Process of Osseointegration

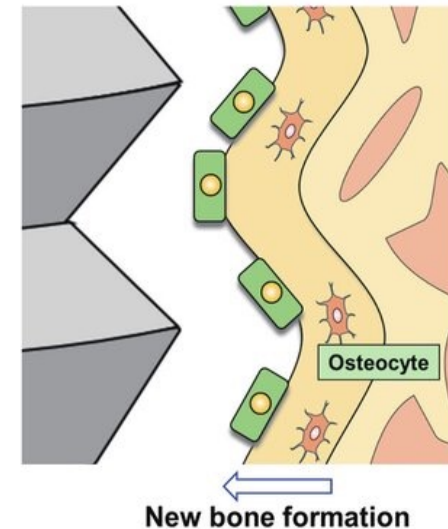


- Once started, osseointegration follows 3 stages:
 1. Implant incorporation via woven bone
 2. Bone mass adaptation to load
 3. Bone remodeling
- Two types of bone regeneration:
 1. **Contact osteogenesis:** from the implant surface towards the bone
 2. **Distant osteogenesis:** from the bone surface toward the implant
- By the end of the 12th week: establishment of **mature lamellar bone connection** with titanium surface

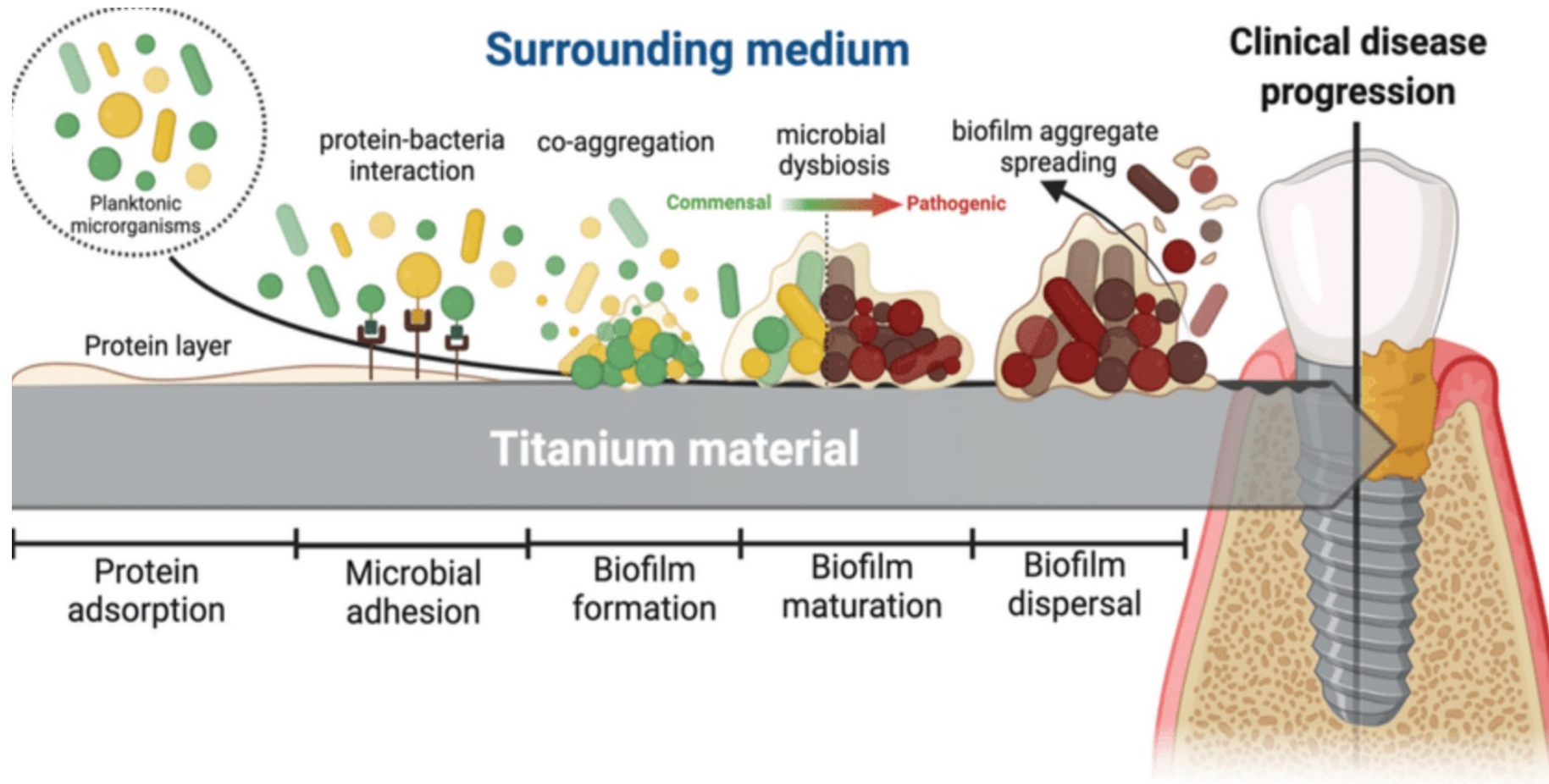
Contact osteogenesis



Distance osteogenesis



Biofilm Formation at the Implant-Tissue Surface



Behavioral Objective #2

Explain how surface energy, topography, roughness, and wettability influence osseointegration, peri-implant tissue health, and biofilm adhesion.

Olivia Prejeant & Savannah King

Search Strategy



P: Patients with dental implants

I: Changes in surface energy, topography, roughness, and wettability on implant surface

C: No modifications to implant surface

O: Effects on osseointegration, peri-implant tissue health, and biofilm adhesion

Databases Used

PubMed

PubMed search string

"Surface energy and Osseointegration"

"Topography and Biofilm Adhesion"

"Roughness and Peri-implant tissue health"

"Wettability and osseointegration"

Search results

102 results, 6 used

Highest level of evidence

Systematic Review and Meta Analysis

Overview of Key Terms



Surface Energy

How a solid surface interacts with liquids and proteins. Measured *indirectly* by contact angles



Topography

3D shape and pattern of an implant over multiple scales like nano, micro, and macro including the grooves, pits, and ridges



Roughness

How much of a surface is irregular. Commonly used in the Ra and Sa values and mainly refers to micro/meso scale texture



Wettability

How readily a liquid spreads on a surface. Measured by contact angle.
Hydrophilic = low angle,
hydrophobic = high angle.



Osseointegration



↑ Protein adsorption



↑ Surface area
↑ Mineral deposition
↑ Tissue regeneration
& new bone formation



Ideal roughness
unclear
1-2 $\mu\text{m Ra}$ - ↑ bone cell
attachment and
differentiation

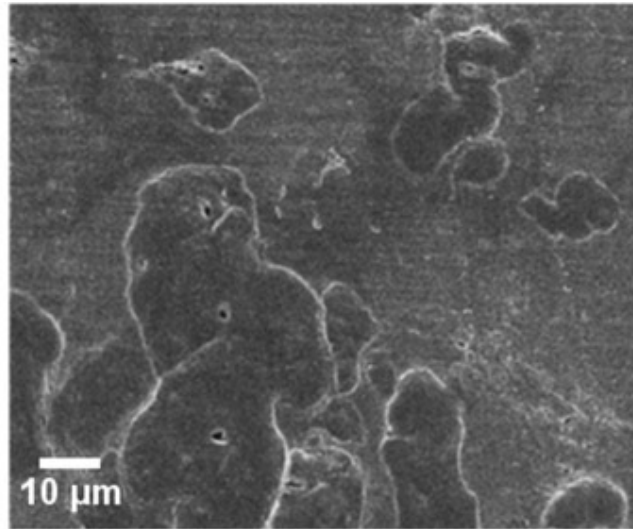


↑ Bone formation
↑ Protein adsorption &
osteoblast attachment
↑ Bone healing time

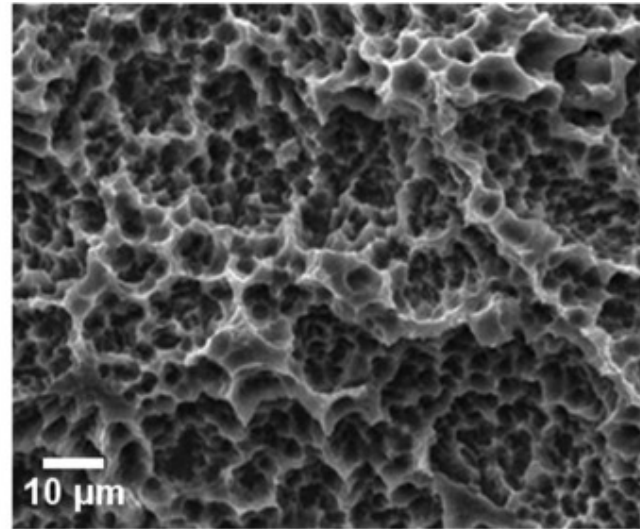




Micro/Macroscale
1 – 100 μm

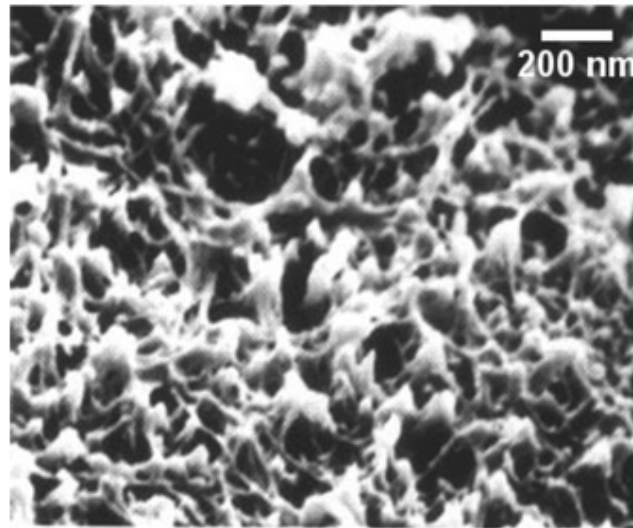


Osteoclast-treated Bone Wafer

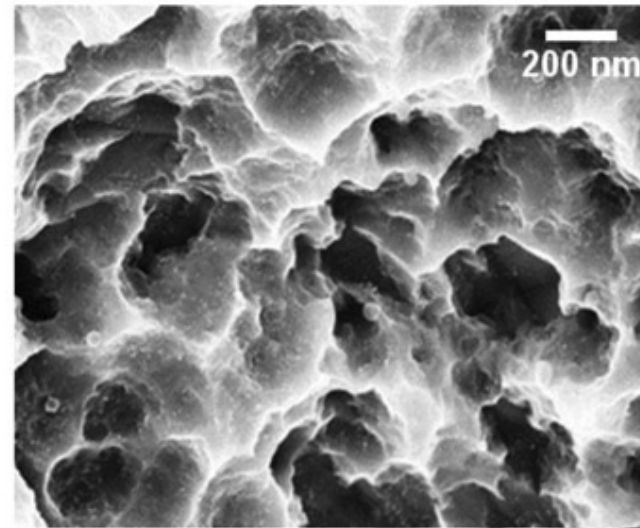


SLA

Meso/Microscale
<1–3 μm



Resorbed Bone Surface



SLA



Peri-Implant Tissue Health



↑ Soft tissue adhesion



↑ Fibroblast & epithelial cell attachment

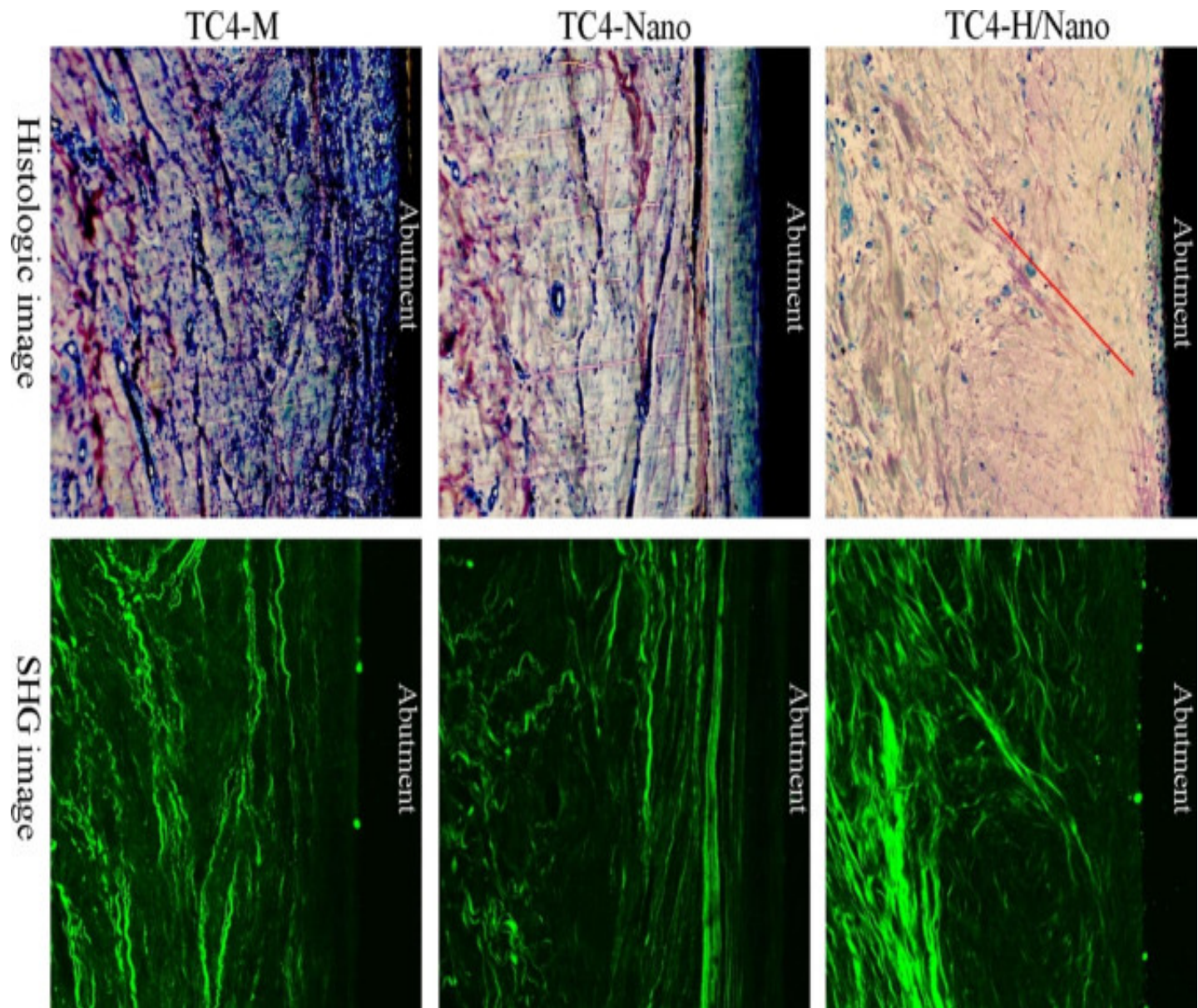


Traps plaque & causes inflammation
Marginal bone loss
Compromised seal



↑ Blood clot formation
↑ Soft tissue adhesion





(Wang et al., 2023)



Biofilm Adhesion



↑ Bacteria



Antimicrobial effect
Biofilm accumulation



↑ Bacterial retention



↑ Bacterial adhesion
Early cell adhesion,
proliferation,
differentiation



Behavioral Objective #3

Differentiate the surface modifications of titanium implants by explaining their influence on bone healing, cell attachment, and microbial colonization.

Dorna Mansouri and Makayla West

Search Strategy

- P:** Patients receiving titanium implants
- I:** Surface modifications of titanium implants
- C:** Machined titanium
- O:** Influence on bone healing, cell attachment, and microbial colonization

Databases used

PubMed

Keywords

"dental implants", "surface modifications", "osseointegration", "cell attachment", "microbial colonization"

PubMed search string

((dental implants) AND (surface modifications)) AND (osseointegration) OR (cell adhesion)

Search results

578 Results, 2 used

Highest level of evidence

Systematic Review and Meta Analysis



Surface modifications of titanium implants



**Machine
sandblasted**



Acid-etched



SLA



**Hydroxyapatite-
coated**



Anodization



Nanostructured



Micro-topographical surface change



**Machine
sandblasted**

HOW

With **alumina particles** (75-500 μm)

WHY

Mimics the resorption pits created by osteoclasts in natural bone

OUTCOME

Negative static charges → **selective activation of osteoblasts & the inhibition of fibroblasts**

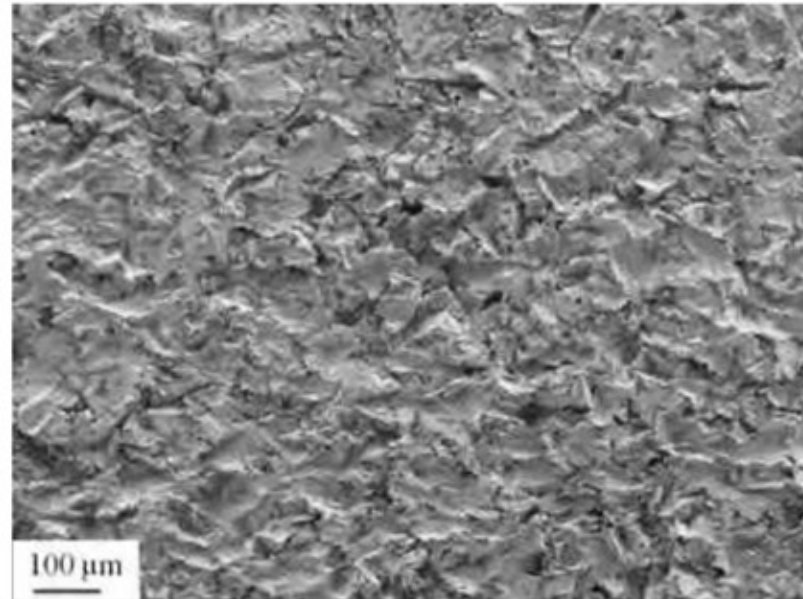
7% (van Velzen et. al. 2014) 1.8% (Bruser et. al. 2012) of sandblasted implants had **peri-implantitis within 10 yrs** compared to 24.8% (Roos-Jansåker et al. 2006) and 12.4% (Fransson et. al. 2005) of implants that were not sandblasted.



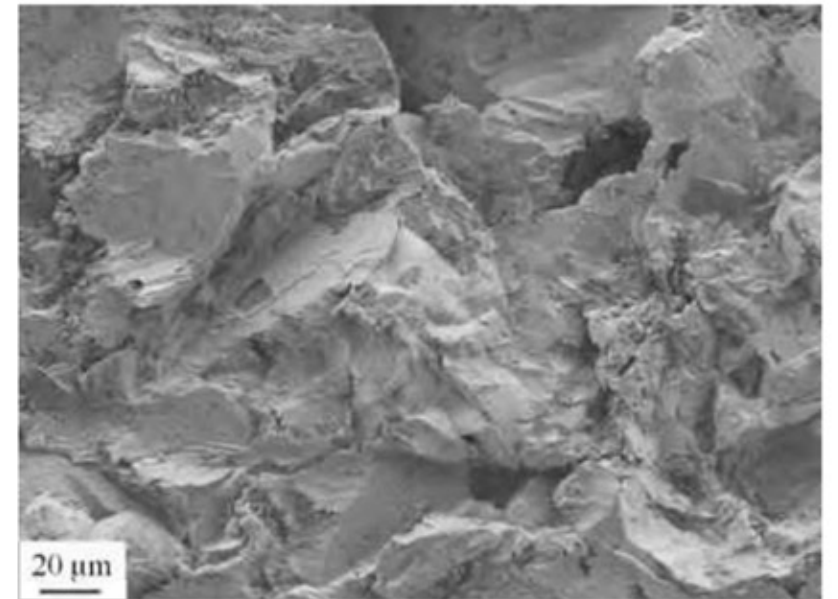
Micro-topographical surface change



**Machine
sandblasted**



(b)



(c)



Surface modifications of titanium implants



**Machine
sandblasted**



Acid-etched



SLA



**Hydroxyapatite-
coated**



Anodization



Nanostructured



Micro-topographical surface change



Acid-etched

HOW

Apply with hydrochloric, sulfuric, nitric acid

WHY

↑ **Surface area** for implant-to-bone connection

OUTCOME

↑ Protein adsorption & hydrophilicity

↑ Bone cell adhesions & growth

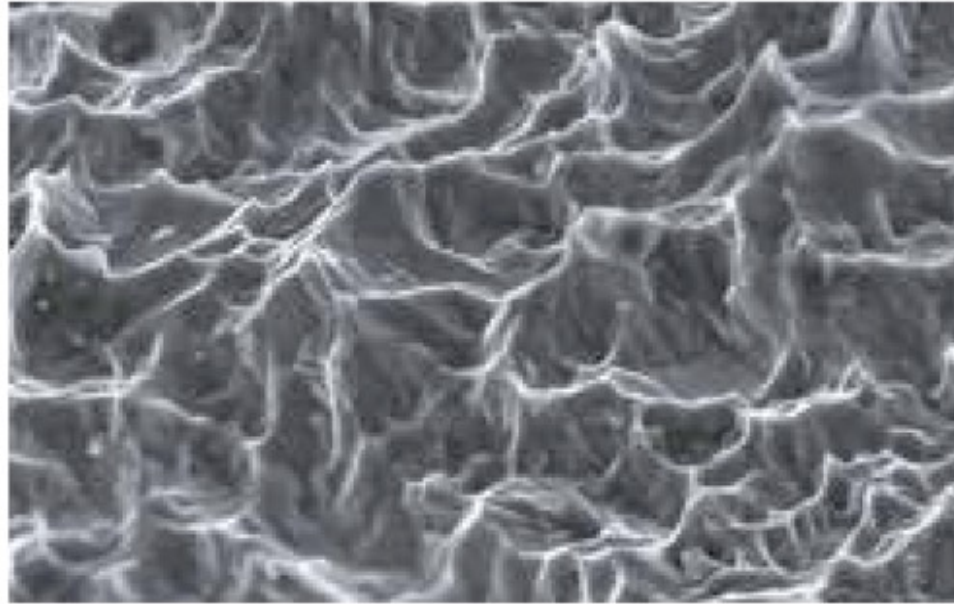
Faster & more stable osseointegration & ST attachment



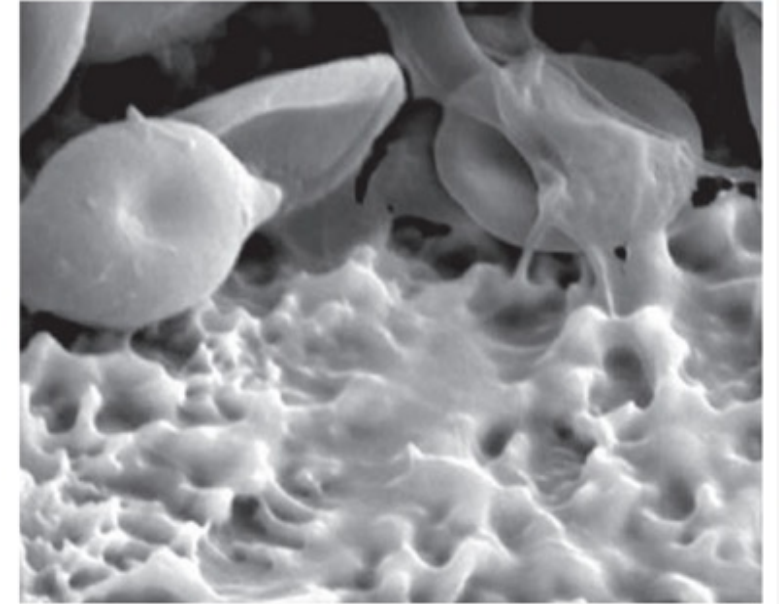
Micro-topographical surface change



Acid-etched



(b)



(c)



Surface modifications of titanium implants



**Machine
sandblasted**



Acid-etched



SLA



**Hydroxyapatite-
coated**



Anodization



Nanostructured



Straumann trademarked technique



SLA

HOW

Large-grit sandblasting + acid-etching

WHY

Combines topography benefits of sandblasting & acid-etching

OUTCOME

↓ **Healing time**

Embedded material may hinder osseointegration

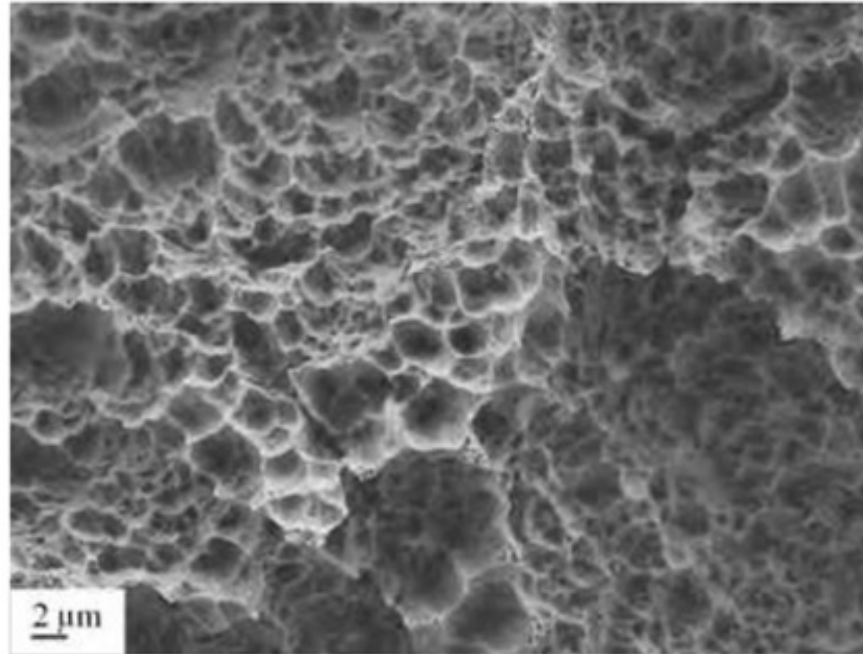
Hydrophobic surface disturbs initial cell attachment



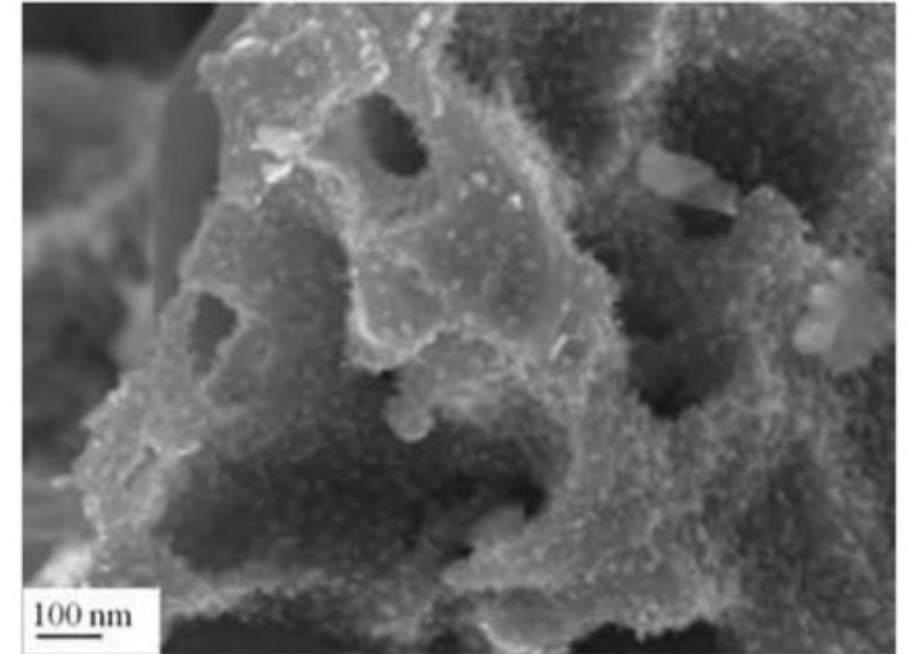
Straumann trademarked technique



SLA



(b)



(c)

Surface modifications of titanium implants



**Machine
sandblasted**



Acid-etched



SLA



**Hydroxyapatite-
coated**



Anodization



Nanostructured



Bioactive coating



**Hydroxyapatite-
coated**

HOW

Electrophoretic, Sol-gel, Hydro-processing, or Biometric

WHY

Mimics bone mineral

OUTCOME

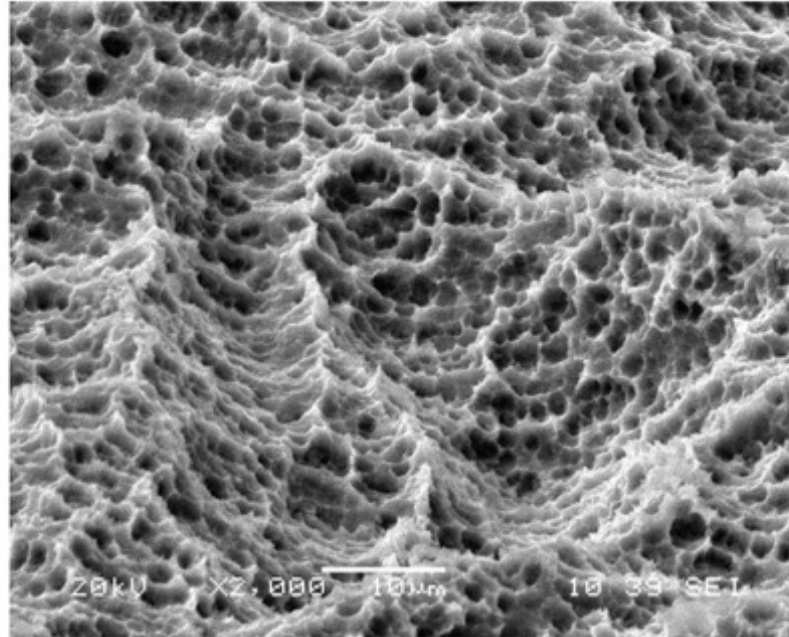
No sinter needed for HA (amorphous/crystalline)
↑ Osseointegration, blood clot retention



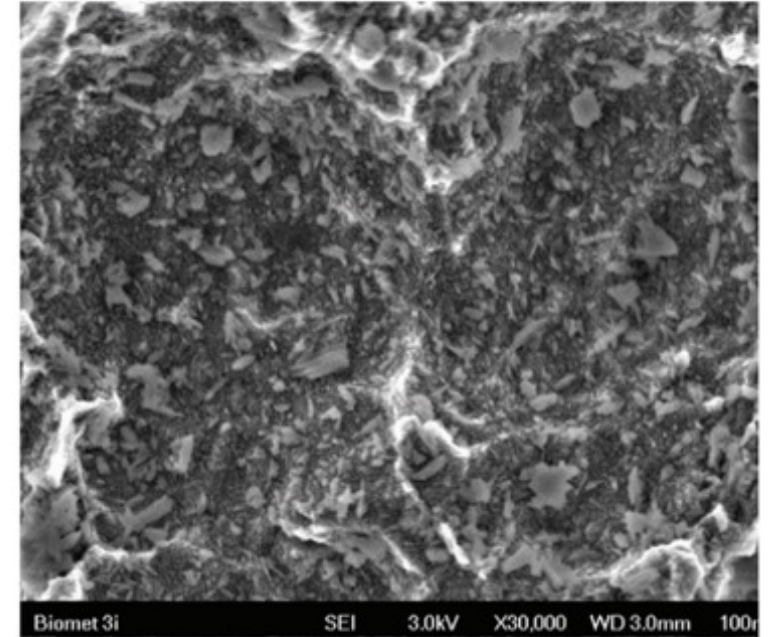
Bioactive coating



**Hydroxyapatite-
coated**



(b)



(c)

Surface modifications of titanium implants



**Machine
sandblasted**



Acid-etched



SLA



**Hydroxyapatite-
coated**



Anodization



Nanostructured



Electrochemical modification



Anodization

HOW

Applying constant **current** or constant **voltage** in an electrolyte containing **strong acids**

WHY

Strong acids → localized **dissolution** of oxide layer
Formation of **micro- & nano-pores**

OUTCOME

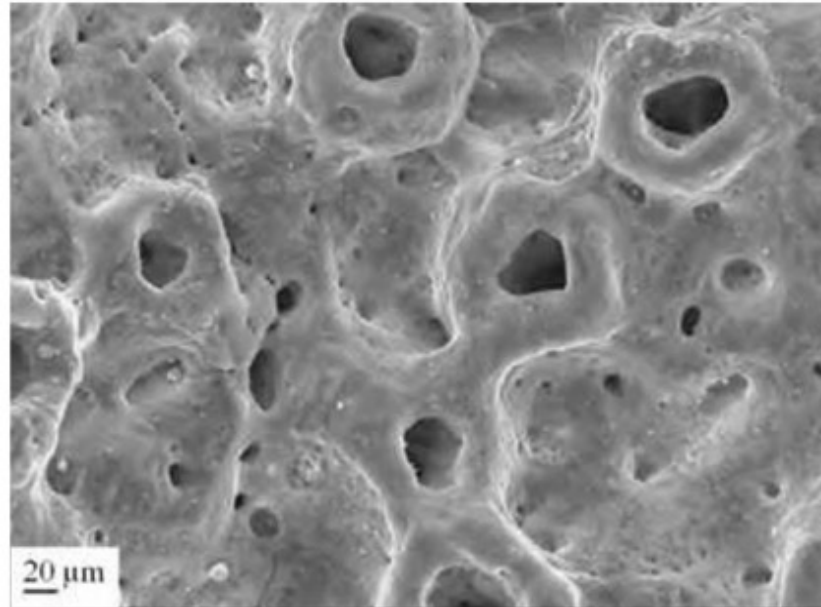
↑ Surface properties
↑ Biological and mechanical performance



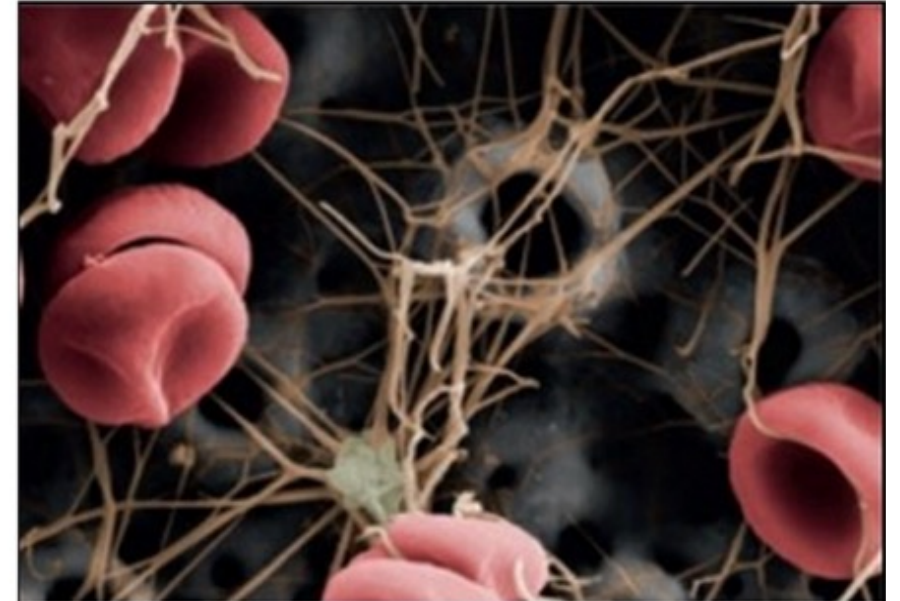
Electrochemical modification



Anodization



(b)



(c)



Surface modifications of titanium implants



**Machine
sandblasted**



Acid-etched



SLA



**Hydroxyapatite-
coated**



Anodization



Nanostructured



Physiochemical approach



Nanostructured

HOW

Anodization, acid-etching/alkali treatment, sol-gel coating, plasma spraying/physical vapor deposition, or laser ablation

WHY

- ↑ Surface area & energy
- Topical cues guide cell behavior & differentiation
- ↑ Osseointegration
- ↑ Corrosion resistance & bioactivity

OUTCOME

- ↑ Osteoblast adhesion, proliferation, differentiation
- ↑ Bone formation & implant stability
- ↑ Long-term implant integration & success
- ↓ Bacterial colonization



Surface modifications of titanium implants



**Machine
sandblasted**



Acid-etched



SLA



**Hydroxyapatite-
coated**



Anodization



Nanostructured



Behavioral Objective #4

Compare and contrast the clinical outcomes and limitations of implant surface innovations based on scientific evidence, focusing on osseointegration rates, peri-implant stability, and risk of peri-implantitis.

Sarah Hsu and Conner Peterson

Search Strategy



P: Patients who received titanium implant treatment

I: Any implant surface modification intended to alter osseointegration or biofilm interactions

C: Conventional/smooth (turned) titanium, other standard micro-rough surfaces, or alternative surface innovation

O: Implant stability and incidence of peri-implantitis

Databases used

PubMed

ChatGPT

Keywords

Dental Implants, Surface Modifications

PubMed search string

((Dental Implants) AND (Surface Modifications)) AND (Stability)

Search results

21 Results, 4 were used

Highest level of evidence

Systematic Review and Meta Analysis



Hydrophilic vs. Standard SAE Implants



Sand blasted, acid-etched (SAE)

- Combining the two techniques = macro- & micro-roughness
- Synergistically optimize biological response

Together, they promote osseointegration, implant stability, and survival rates.

Increased wettability (hydrophilicity) promotes bone healing & early osseointegration.



Hydrophilic vs. Standard SAE Implants



- Mod-SAE appeared to **improve the first phases of osseointegration** in pre-clinical studies (in vivo, in vitro, animal studies)
- One study suggested that **hydrophilic** surfaces implants **osseointegrate faster** and **gain stability 2.24-fold faster** than SAE implants.
- Another study reported that implants with a hydrophilic surface seem to **avoid the reduction in ISQ values during the remodeling phase**



Hydrophilic vs. Standard SAE Implants



Other studies have report differently

- **No significant differences**
 - Hydrophilic and hydrophobic surfaces
 - Biomechanical properties and the amount of bone to implant contact during the early osseointegration period
- **Similar levels of implant stabilization**
 - Standard SAE and chemically modified SAE implants for patients with type 2 diabetes



Hydrophilic vs. Standard SAE Implants



Comparisons of ISQ values between control and test groups at weeks 0, 3, 6, and 8

- Both groups had relatively high ISQ values at all observation times
- **No statistically significant difference** between the groups at all measurement intervals
- Modification of implant surfaces and increasing hydrophilicity is useful, but there's **not enough research to determine which is better**

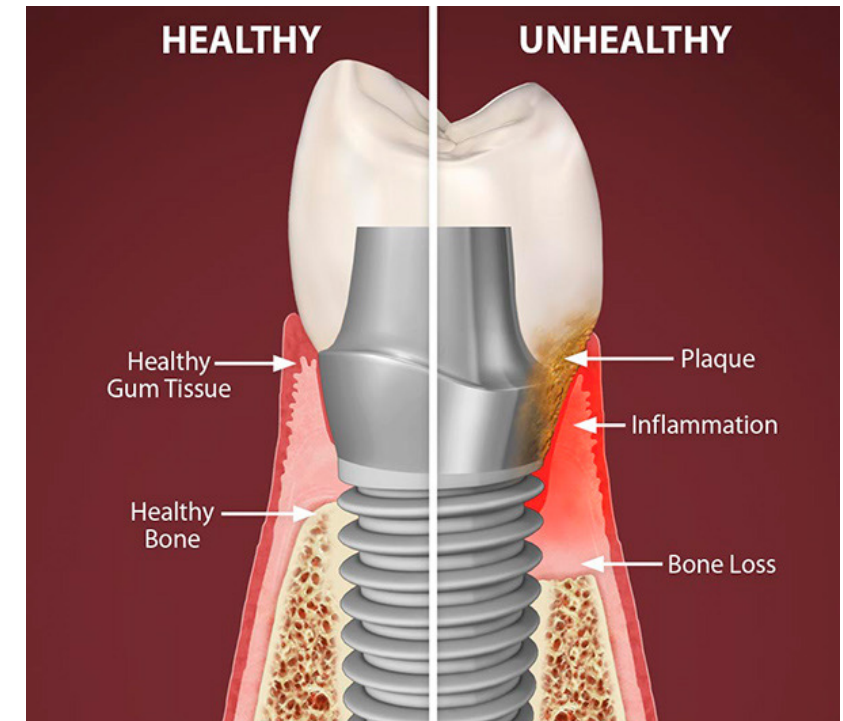
*Implant stability quotient (ISQ)



Peri-implantitis



- Inflammatory disease that compromises peri-implant tissues & supporting bone
- Can lead to implant loss/failure
- Affects ~19.53% of patients & 12.53% of implants
- Main etiological factor is **oral biofilm**, that is affected by the implant surface modifications



Defining Peri-implantitis



- Varies Greatly
 - PD: >6mm or 4-5mm
 - BOP & Suppuration
 - Radiographic Bone Loss
 - >3mm or >1.8mm

TABLE 2 Definitions of periimplantitis.

Study	Definition of periimplantitis
Aghazadeh et al., 2022 (32)	Probing pocket depths of at least 5 mm, presence of bleeding on probing and/or suppuration, radiographic bone loss of 2 mm or more from implant placement to screening, and an angular peri-implant bone defect of 3 mm or greater
Carcuac et al., 2020 (15)	Probing pocket depths of 6 mm or more, presence of bleeding on probing, reduced marginal bone level and progressive bone loss greater than 1 mm post-treatment
Deppe et al., 2007 (18)	Probing pocket depths of at least 5 mm, presence of bleeding on probing, radiographic evidence of progressive vertical bone loss, and clinical signs of inflammation
Jemt et al., 2021 (19)	Bone loss exceeding 0.4 mm, mucosal inflammation, presence of plaque and/or suppuration, and radiographic evidence of marginal bone loss
Khoury et al., 2001 (20)	Bone loss of more than 50% of the implant length, augmented probing depths, bleeding on probing, and radiographic evidence of intrabony defects
La Monaca et al., 2018 (21)	Progressive bone loss of 3 mm or more detected on radiographs, the presence of bleeding on probing and/or suppuration, and probing depths of at least 5 mm
La Monaca et al., 2024 (22)	Progressive angular bone loss of at least 3 mm beyond crestal bone level changes, the presence of bleeding on gentle probing and/or suppuration, and implants in function for more than 12 months
Leonhardt et al., 2003 (23)	Marginal bone loss of at least three implant threads compared to baseline radiographs, bleeding on probing and/or suppuration from peri-implant sulci, and microbiological confirmation of peri-implant pathogens
Mercado et al., 2018 (24)	Probing pocket depths exceeding 4 mm, the presence of bleeding on probing and/or suppuration, a minimum radiographic bone loss of 20%, and implants that have been in function for at least 2 years
Noelken et al., 2023 (25)	Probing pocket depths greater than 5 mm, the presence of bleeding on probing and suppuration, and radiographically confirmed bone loss
Roccuzzo et al., 2017 (26)	Probing pocket depths of at least 6 mm, no implant mobility, bleeding on probing and/or suppuration, and radiographic bone loss exceeding three implant threads compared to baseline
Roccuzzo et al., 2020 (27)	Probing pocket depths of 6 mm or greater, the presence of bleeding on probing and/or suppuration, radiographic bone loss beyond crestal changes, and the absence of implant mobility
Roccuzzo et al., 2021 (28)	Probing pocket depths reach or exceed 6 mm, bleeding on probing, radiographic evidence of progressive bone loss, and the presence of pus or inflammation
Romandini et al., 2024 (16)	Probing pocket depths of 6 mm or more, bleeding and/or suppuration on probing, and radiographic evidence of marginal bone loss equal to or greater than 3 mm
Roos-Jansåker et al., 2011 (29)	Radiographic bone loss of at least 1.8 mm following the first year in function, the presence of bleeding and/or pus on probing, and inclusion criteria of non-mobile implants
Roos-Jansåker et al., 2014 (30)	Radiographic bone loss of at least 3 threads (≥ 1.8 mm), the presence of a vertical defect component, and bleeding on probing and/or suppuration
Schwarz et al., 2009 (31)	Probing pocket depths greater than 4 mm, presence of bleeding on probing and/or suppuration, radiographic evidence of bone loss, and an intrabony defect component of at least 3 mm



Peri-implantitis



- Incidence of peri-implantitis seems not to differ between modified and non-modified (i.e., turned) implants in the clinic
- Progression and severity of peri-implantitis appear linked to implant surface properties
 - Pre-clinical in-vivo: **faster disease progression with modified implants**





Peri-implantitis:

Recurrence & treatment failure

- **Recurrence**
 - Modified surfaces: 44% & 24.3%
 - Turned surfaces: More Stable outcomes
- **Treatment failure**
 - Surface roughness is a key determinant of recurrence & long term stability
 - Patient factors: **Smoking & Diabetes**





Peri-implantitis:

Implant loss, resolution, bone loss

- **Implant loss**
 - Roughened Surfaces: TPS 45% vs SLA 20%
 - Turned Surfaces: 27%
 - However: Reconstructive surgical approach decreased implant loss rates
- **Resolution**
 - PD was reduced but amount varied greatly on roughened surfaces
 - 8.9mm → 3.5mm vs 5.05mm → 3.08mm
 - SLA > TPS
- **Bone loss**
 - Regeneration dependent on surface
 - SLA: +2.1mm, +2.7mm, 3.2mm (w/ Autographs)
 - TPS: Less long term stability
 - Smooth surface rarely evaluated



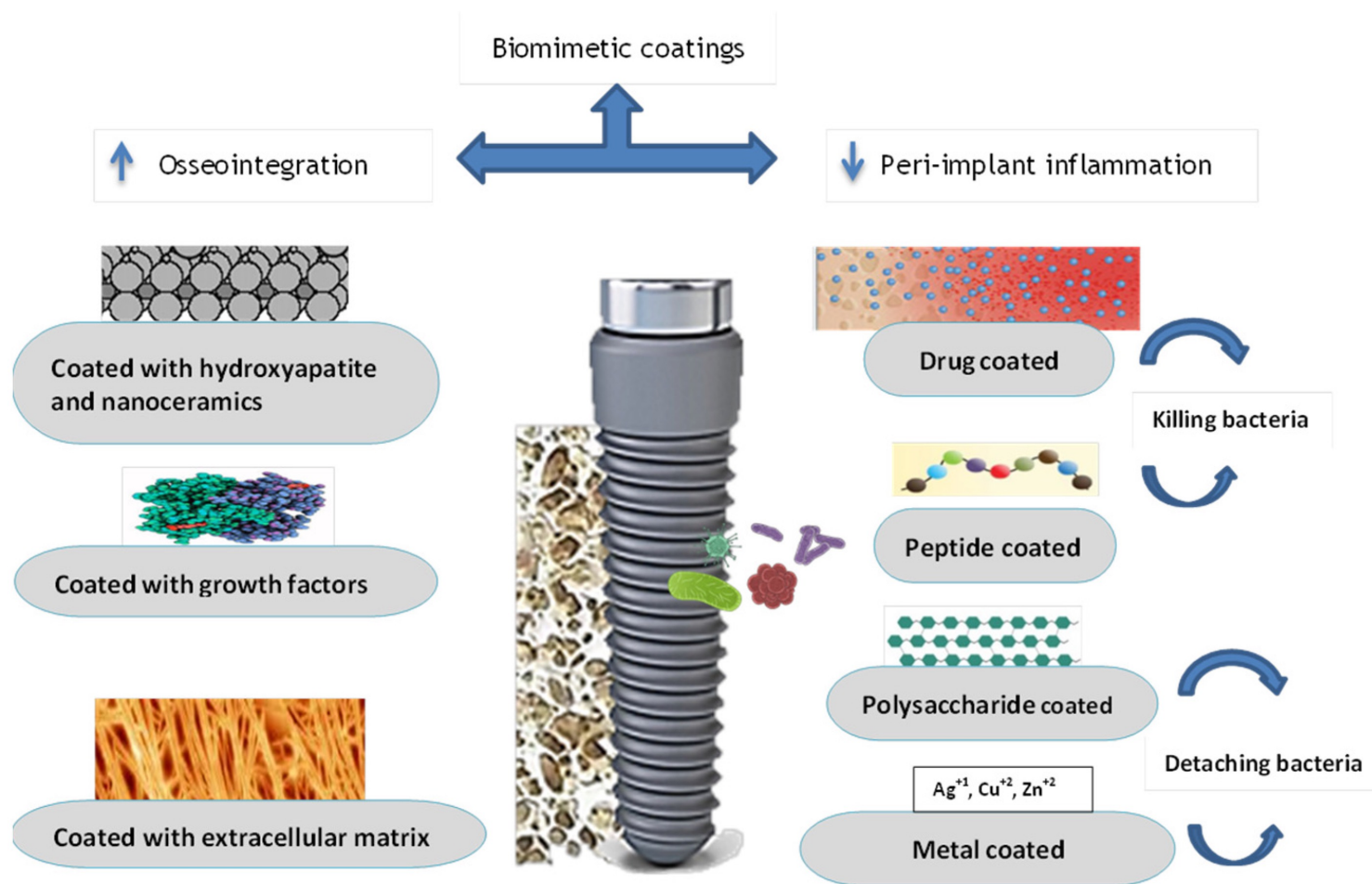
Peri-implantitis risk



- **No significant difference** based on surface modification type
- **Better maintenance, patient selection, prosthetic design, surgical technique, and early detection** likely matter more than whether the implant is rough or machined.



Bioactive surface modifications



Bioactive surface modifications



Osseointegration: Statistically significant improvement in BIC for surface modified implants in comparison to unmodified implants

Longevity: Statistically significant improvement

PROBLEMS: Success varies greatly

- Bioactive agent used, concentrations of coating, patient differences, expensive!



Behavioral Objective #5

Apply a biologically informed decision tree to select implant surfaces and adjunctive treatments tailored to patient-specific factors such as bone quality, systemic health, microbial risk, and esthetic demands.

Tahj Boyce

Search Strategy

P: Patients receiving dental implants with varied bone and health profiles.

I: Biologically tailored implant surfaces and adjunctive treatments

C: Standard implant surfaces and protocols

O: Improved osseointegration, bone stability, and tissue health

Databases used

PubMed

Google

Keywords

Dental implants, osseointegration, implant surface modification, hydrophilic surface, hydroxyapatite coating, surface roughness, smooth collar, bone density, periodontitis, bioactive surface

PubMed search string

Search string

Search results

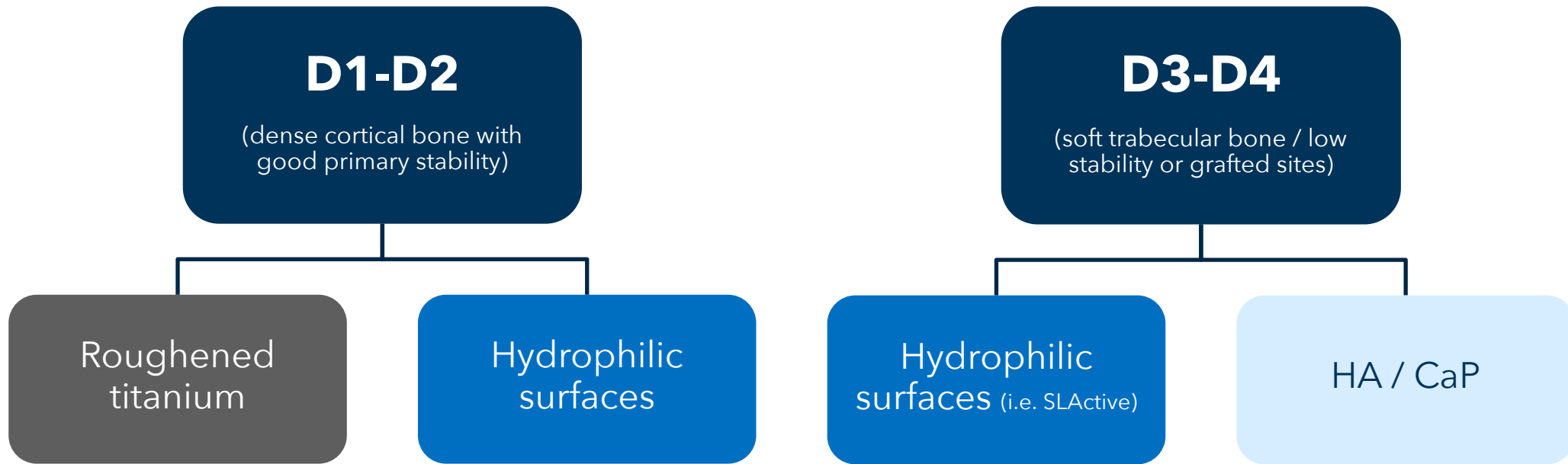
5207Results, 7 used

Highest level of evidence

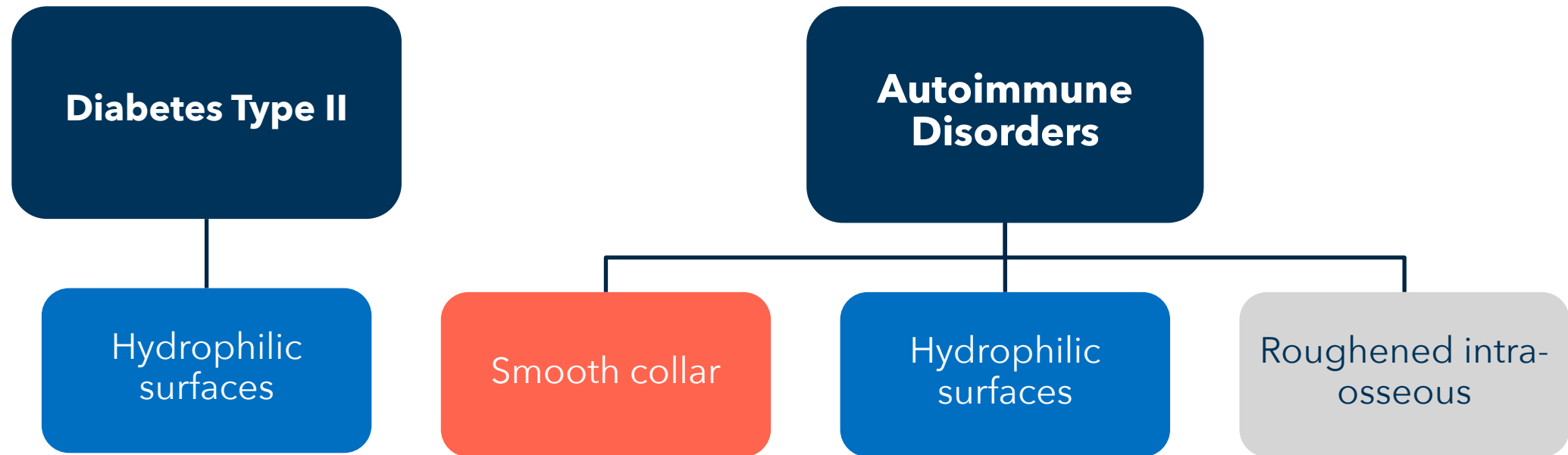
Systematic Review and Meta Analysis



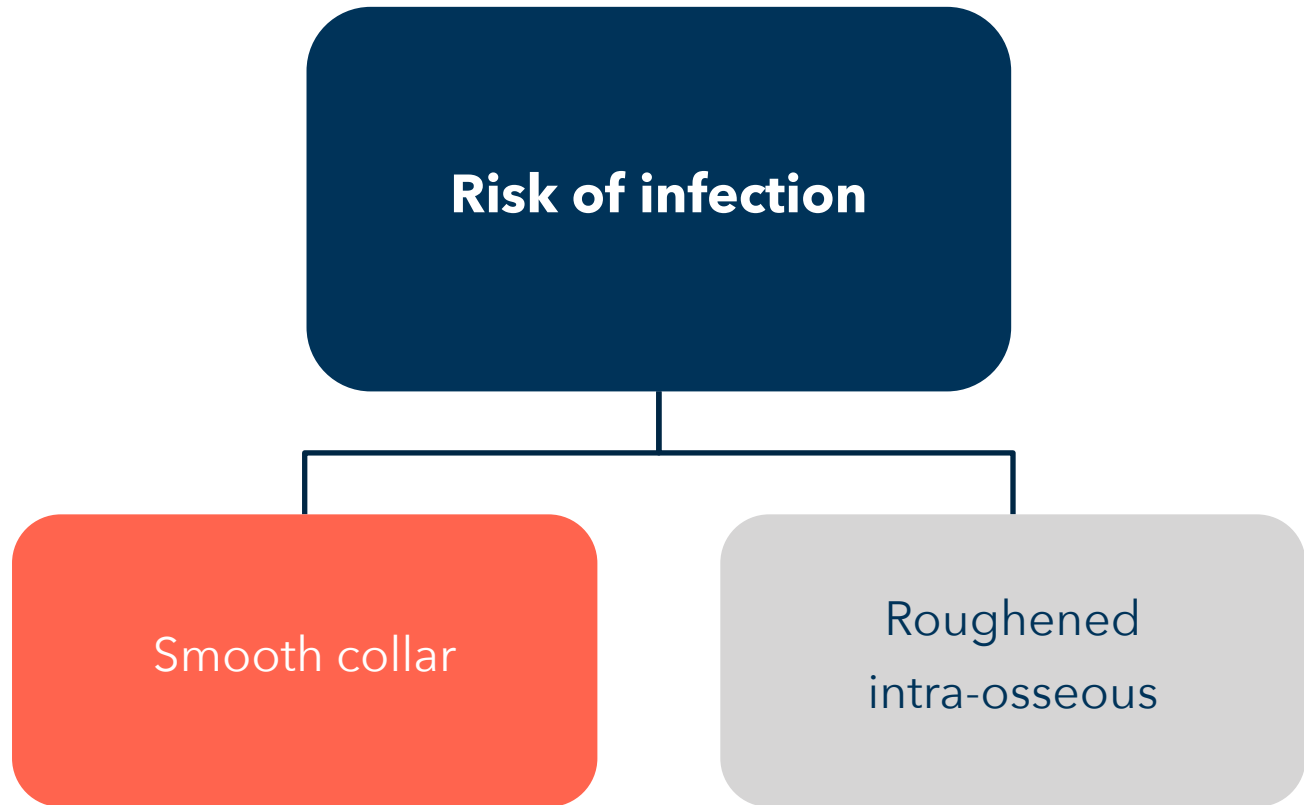
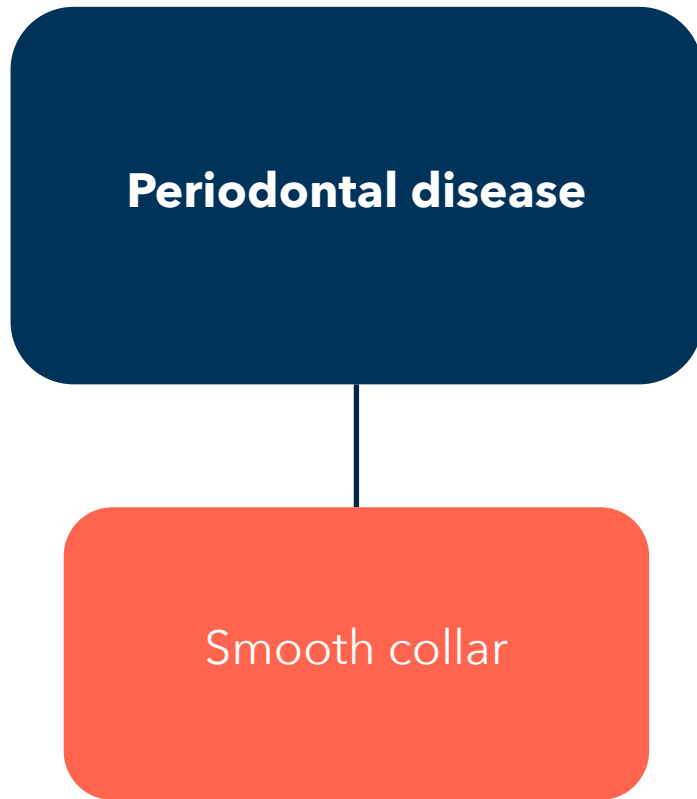
Bone Density



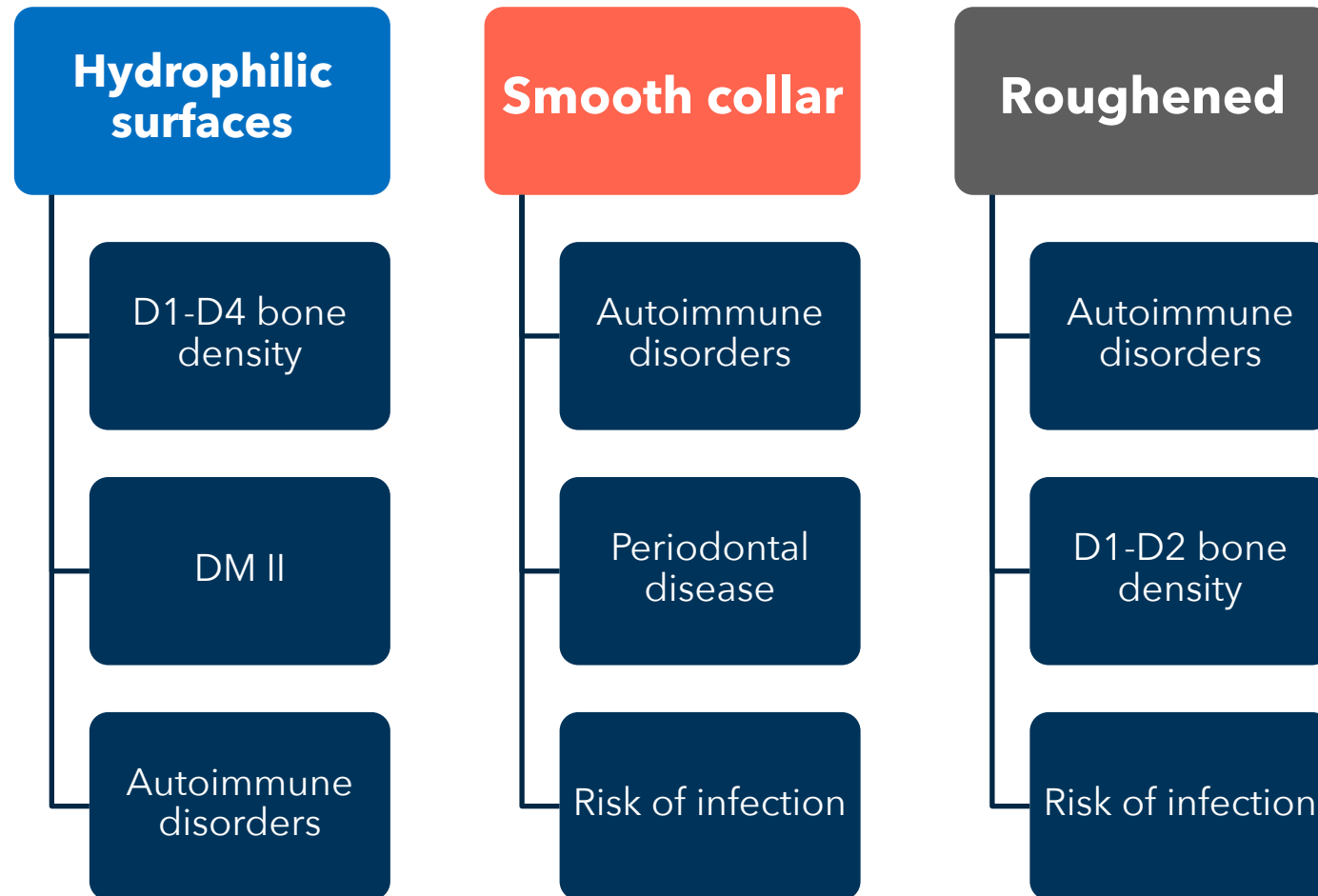
Systemic Diseases



Microbial Risk



Summary



Clinical Cases

Clinical Case #1

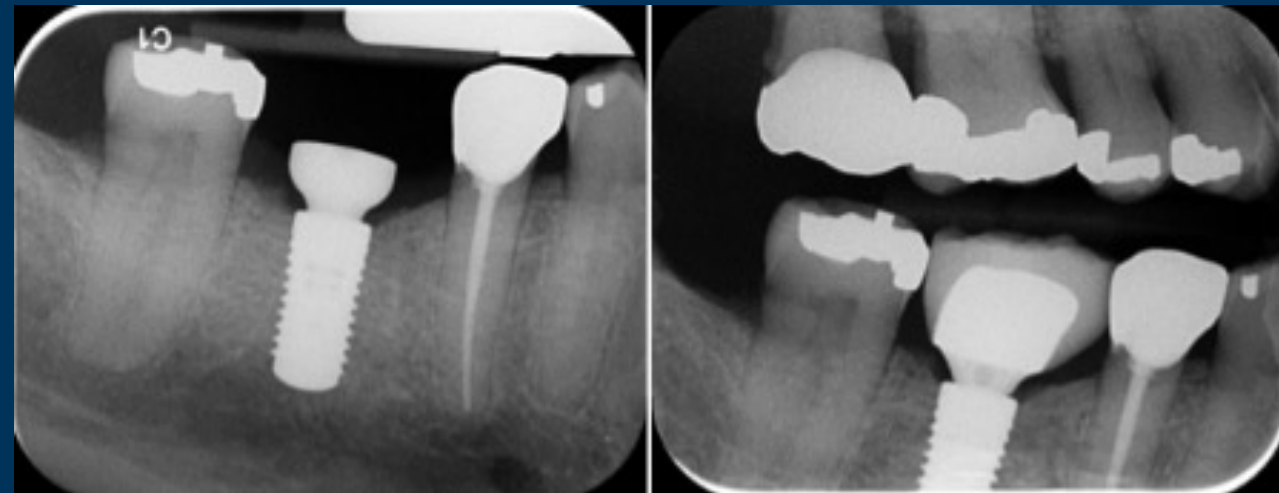
58 yo Caucasian Female

Site: #30

Med hx: HTN, DMII, Depression

Implant used: Straumann BL

Results: implant placed and restored in 2022 – peri-implant health achieved and in good function.



Clinical Case #2

59 yo Caucasian Female

Site: #26

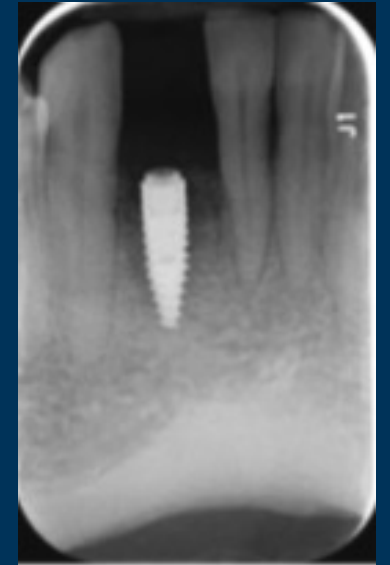
Med hx: HTN, Hyperlipidemia, H/o Stroke, seizures, Joint replacement.

PSH: Brain surgery in 1988 to control seizures and hemiparesis.

History of **severe chronic perio** (Stage 3 Grade C)

Implant used: Straumann BL taper

Result: **failed due to poor osseointegration**



Summary

BO #1

- Osseointegration depends on the cell interactions at the implant-tissue interface.
 - Platelets and immune cells (neutrophils and macrophages)
 - Osteoprogenitor cells
- Titanium implants have bacteriostatic properties
 - Dysbiosis → biofilm accumulation → inflammation and peri-implantitis.
- Increased surface roughness
 - Enhance osseointegration
 - Increase risk of biofilm accumulation

Summary

BO #2

Osseointegration

- Surface energy and wettability are similar where it helps increase protein adsorption
- Topography increases surface area, mineral deposition, and integration
- Ideal roughness is unclear

Peri-implant tissue health

- Surface energy and wettability inc. soft tissue adhesion
- Topography helps with fibroblast and epithelial attachment
- Too much roughness can result in a marginal bone loss = not a sealed implant

Biofilm Adhesion

- Topography can have an antibacterial effect
- Rough implants will increase bacterial retention
- A high wettability will increase the bacterial adhesion
- High surface energy attracts more bacteria

Summary

BO #3

The most effective bioactive coating for titanium dental implants is **hydroxyapatite**.

- ↑ Osseointegration
- Highly biocompatible
- Precise control over the surface topography
- Blood clots are well-retained over the threaded part of implant

Machine Sandblasted:

- Increased surface area for bone contact
- Supports mechanical interlocking
- Osteoblasts are selectively activated, while fibroblasts are inhibited

Acid-Etched:

- Support mechanical inter-locking
- Increased protein adsorption and hydrophilicity --> increased bone cell adhesion and growth --> faster and more stable osseointegration and soft tissue attachment

Summary

BO #3

SLA:

- large-grit sandblasting --> generates a macro-roughness followed by acid-etching --> superposes a micro-roughness
- Reduced unloaded healing time from 12 to 6 weeks
- Gold standard for developing novel implants

Anodization:

- Strong acids in the electrolyte causes localized dissolution of the oxide layer along current convection lines while allowing it to grow in other regions
- Formation of micro- and nano-pores on the titanium surface.
- Thickened, porous titanium oxide layer with modified microstructure and crystallinity,
 - enhanced implant surface properties and potential for improved biological and mechanical performance

Nano-structured:

- Increased surface area and energy, promoting better protein adsorption and osteoblast attachment
 - Improves osseointegration by enhancing bone-to-implant contact at the nanoscale
 - Reduced bacterial colonization due to unfavorable nanostructured patterns.
- Improved long-term implant integration and clinical success rates

Summary

BO #4

- Modification of implant surfaces and increasing hydrophilicity is useful, but there's not enough research to determine which is better.
- Rough surfaces may reduce early failures and improve integration but may require more vigilance for soft tissue health and hygiene.

Summary

BO #5

- Choosing the right implant surface means matching the material to the patient's biology.
- Overall, hydrophilic surfaces are the most adaptable because they promote healing in a variety of conditions, while smooth collars remain key for long-term health and maintenance.

References



Six Thinking Hats®



White Hat (Faculty): Provides objective data and scientific evidence about the dental material.



Red Hat (Patient): Expresses emotions, concerns, and preferences regarding the material.



Black Hat (Dentist): Evaluates potential risks, challenges, and clinical implications.



Yellow Hat (Sales Rep): Highlights the benefits and advantages of the material.



Green Hat (Manufacturer): Explores innovation, and future possibilities for the material.



Blue Hat (Board of Dentistry): Ensures all perspectives and standards are considered.

